Data Integrity and Validation -- Guided Notes ANSWER KEY

Unit tp-5.5 | CompTIA Network+ N10-009 Obj. FC0-U71 5.5 | N10-008 FC0-U71 5.5

For instructor use / distribute after students complete the notes.

Question 1

Data integrity means data is _______ (correct), _______ (complete), and _______ (not changed without authorization). Organizations achieve data integrity through _______ (enforcing data types and required fields), _______ (detecting unauthorized changes), and access controls.

Answer: Accurate; complete; unaltered; validation; hashing (checksums).

Question 2

Input validation ensures data entered into a system meets _______ criteria before being processed. Frontend validation in a browser gives _______ feedback but can be bypassed. Server-side validation is _______ because it is enforced in code the user cannot modify. Both should be used together.

Answer: Expected (defined); immediate (user-friendly); more trustworthy (authoritative).

Question 3

A _______ constraint in a database prevents NULL (empty) values in a column. A _______ constraint ensures all values in a column are different. A _______ constraint limits which values are permitted (e.g., grade must be A, B, C, D, or F). A _______ constraint enforces the relationship between two tables using keys.

Answer: NOT NULL; UNIQUE; CHECK; foreign key (referential integrity).

Question 4

A checksum or _______ (hash) function produces a fixed-length value from input data. If even one _______ in the original data changes, the hash changes completely. This property allows detection of _______ during transmission or storage. Common hash functions include _______ and _______.

Answer: Hash; bit; tampering (corruption); SHA-256, MD5 (note: MD5 is deprecated for security use).

Question 5

Data _______ is the process of ensuring data is formatted correctly for analysis (fixing errors, removing duplicates, standardizing formats). _______ data (incorrect, inconsistent, or incomplete records) leads to unreliable analysis results. The rule is: _______ in, _______ out.

Answer: Cleaning (sanitization); dirty; garbage; garbage.

Question 6

Referential integrity ensures that a foreign key value in one table must match a _______ key that _______ in the referenced table, or be NULL. If a parent record is deleted, the DBMS can _______ delete all child records (CASCADE), or _______ the deletion if child records exist (RESTRICT).

Answer: Primary; exists; cascade; block.

Question 7

Data ________ involves documenting data assets -- what data exists, where it is stored, who owns it, and its ________ level. Strong governance practices include ________ policies that specify how long data must be retained and when it must be ________. Retaining data longer than necessary increases ________ risk.

Answer: Governance; classification (sensitivity); retention; destroyed (deleted); breach.

Question 8 -- Real World Application

REAL WORLD: A student builds a web form that accepts a user's age as a text field. A malicious user enters 'abc' as their age, which crashes the database query. Another user enters -5 as their age, which the system accepts. Identify both problems and describe the server-side validation rules needed to fix them.

Answer: Problem 1 -- Type validation failure: the input 'abc' is not a number, but the application passes it to the database query without checking. The fix is: validate that the input is a positive integer before processing. In server code: check that the value can be parsed as an integer (not a string or float) -- reject anything that fails type conversion with an error message. Problem 2 -- Range validation failure: -5 is a valid integer but an impossible age. The fix is: apply a range constraint -- age must be >= 0 and <= 120 (or whatever reasonable maximum). Return an error if the value is outside this range. Combined server-side rule: input must be a whole number AND be between 0 and 120. Reject any input that fails either check and return a clear user-facing error message. Never trust client-side validation alone -- browsers can be bypassed.